Recoil and explosions

antocciphysics

UNIT 3 / DAY 5 / HONORS

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Use momentum conservation for an isolated system.

SUCCESS CRITERIA - I can...

1. Choose a system and check external impulse.
2. Add momentum with signs.
3. Explain equal-and-opposite momentum changes.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

01 Bring back an earlier idea

A cart has momentum -15 kg·m/s and mass 3 kg. Find velocity and its new momentum if velocity reverses at unchanged speed.

Marzano check - circle: ① ② ③

02 Explain a worked example

Reference: $p_{total,\,i} = p_{total,\,f}$.
Negligible net external impulse. Internal forces can change each part's momentum but not the system total.

Worked reasoning: Two carts initially at rest push apart with negligible external impulse. A 1 kg cart leaves at +4 m/s; a 2 kg cart must have momentum -4 kg·m/s, hence velocity -2 m/s.

Your turn: Draw before-and-after momentum arrows for the two-cart system.

Marzano check - circle: ① ② ③

03 Represent and interpret

Two carts push apart from rest on an ideal track. Draw their momentum arrows. How must the two momenta compare?



A moves left B moves right

Marzano check - circle: ① ② ③

04 Solve and justify

A 60 kg person and 3 kg tool are initially at rest relative to a spacecraft. The tool leaves at +8 m/s in that frame. Find person velocity, neglecting external impulse.

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

Recoil and explosions

antocciphysics

Continue down each column.

05 Analyze the reasoning

Why can the two parts gain kinetic energy during a release while total momentum remains zero?

Marzano check - circle: 1 2 3

06 Choose a method

A projectile’s velocity is (12, -16) m/s. Find its speed. What information would locate it?

Marzano check - circle: 1 2 3

07 Exit check

Two fragments of masses 2 and 3 kg separate from rest. The 2 kg fragment moves at +9 m/s. Find the other velocity and total kinetic energy.

Marzano check - circle: 1 2 3

08 STRETCH: Apply it another way

A 60 kg person and 10 kg board start at rest. After a push, the board moves at +3 m/s relative to the ground. Find the person's ground velocity and the board's speed relative to the person.

Marzano check - circle: 1 2 3